Investment Analysis Midterm - Short Answer Quiz

1. **[Ch.2]** What defines a money market instrument? Explain why a Treasury note is not one, and name two money market instruments.

2. **[Ch.2]** In a repurchase agreement (repo), what does the dealer do, and why is a repo considered a very safe short-term loan?

3. **[Ch.2]** What are Eurodollars? Does the prefix "Euro" mean that the deposit must be held in Europe?

4. **[Ch.2]** What is a mortgage-backed pass-through security, and how does securitization shift risk?

5. **[Ch.2]** Which two characteristics define common stock as a claim on the firm, and what does each mean?

6. **[Ch.2]** Why is preferred stock called a hybrid security? What happens if the firm skips a preferred dividend?

7. **[Ch.2]** What is an American Depositary Receipt (ADR), and what is its advantage for a U.S. investor?

8. **[Ch.2]** A stock in a price-weighted index splits 2-for-1. Why must the divisor be changed, and what happens to that stock's weight in the index?

9. **[Ch.2]** What does an equally weighted index imply for an investor's portfolio, and why does maintaining it require trading?

10. **[Ch.2]** What rights does the holder of a call option have, what is the holder's maximum loss, and what happens if the asset price ends below the exercise price?

11. **[Ch.2]** When is a put option in the money, and why is being in the money not the same as being profitable for the buyer?

12. **[Ch.2]** Distinguish a long futures position from a long call position in terms of right versus obligation, up-front payment and potential loss.

13. **[Ch.2]** At what point is an investor indifferent between a taxable bond and a municipal bond, and why are municipal bonds mostly held by high-bracket investors?

14. **[Ch.2]** State the two flaws of the bank discount yield as a measure of the return on a Treasury bill, and how the bond equivalent yield corrects them.

15. **[Ch.2]** For the same Treasury bill, rank the bank discount yield, the bond equivalent yield and the effective annual yield from lowest to highest, and explain what drives each step.

16. **[Ch.2]** What does it mean for a money market fund to "break the buck," and what triggered it for the Reserve Primary Fund in 2008?

17. **[Ch.2]** How do TIPS protect an investor against inflation?

18. **[Ch.2]** Distinguish general obligation bonds from revenue bonds. Which is riskier in terms of default?

19. **[Ch.2]** Compare price-weighted, value-weighted and equally weighted indexes. Which kind of stock dominates the first two?

20. **[Ch.3]** In which market do the proceeds of a sale go to the issuing firm, and why does the secondary market still matter to firms?

21. **[Ch.3]** Distinguish firm commitment from best efforts underwriting. Who bears the risk of an unsold issue in each?

22. **[Ch.3]** What does the evidence say about average first-day and long-run IPO returns, and what explains the first-day pattern?

23. **[Ch.3]** A dealer quotes a bid of $34.75 and an asked price of $34.90. At which price does an investor buy, at which does she sell, and what does the spread represent?

24. **[Ch.3]** Compare a market order and a limit order in terms of certainty of execution and certainty of price.

25. **[Ch.3]** A stock trades at $50 and an investor places a limit buy order at $47. What happens to the order, and how does it relate to market liquidity?

26. **[Ch.3]** What does a stop-loss sell order at $55 do for an investor who holds a stock trading at $60, and what does it not guarantee?

27. **[Ch.3]** What is high-frequency trading, and what do proponents and critics argue about its effect on liquidity?

28. **[Ch.3]** Explain how buying on margin affects an investor's return and risk, and who sets the initial margin requirement.

29. **[Ch.3]** When does a margin call occur for a margin-purchased (long) position and for a short sale, and why do the two differ?

30. **[Ch.3]** Describe a short sale: what does the seller do, when does the position profit, and how do the maximum gain and loss compare?

31. **[Ch.3]** In a short sale, what happens to the sale proceeds, what margin is required, and who receives the dividends paid on the borrowed shares?

32. **[Ch.3]** How do ECNs and algorithmic (high-frequency) trading change the way securities are traded, and what are the benefits of electronic trading?

33. **[Ch.3]** Name the four pillars of securities regulation and state what each does.

34. **[Ch.4]** What exactly does an investor who owns 2% of a mutual fund's shares own, and what is not guaranteed?

35. **[Ch.4]** Name the four core services an investment company provides to small investors, and say whether a return above the market is one of them.

36. **[Ch.4]** How is a fund's net asset value (NAV) computed, which items are deducted, and why does issuing new shares at NAV leave NAV unchanged?

37. **[Ch.4]** At what price is an order to buy open-end mutual fund shares at 10:00 a.m. filled, and how do closed-end funds and ETFs differ?

38. **[Ch.4]** Describe a closed-end fund: how its shares are issued, who trades with whom, and how its market price relates to NAV.

39. **[Ch.4]** What is a unit investment trust (UIT), and why are its management fees very low?

40. **[Ch.4]** How does a hedge fund differ from a mutual fund in who may invest, regulation, strategies, fees and redemptions?

41. **[Ch.4]** Distinguish a balanced fund from an asset allocation fund.

42. **[Ch.4]** Why do investors tend to underestimate a fund's expense ratio, and is the reported fund return before or after it?

43. **[Ch.4]** What is a 12b-1 fee, how large can it be, and why is it economically like a sales load?

44. **[Ch.4]** What does a back-end load (CDSC) that declines each year do economically, and who receives it?

45. **[Ch.4]** A fund reports an 11% one-year return and an investor paid a 5% front-end load. Does the reported return equal her return?

46. **[Ch.4]** Class A shares charge a front-end load and a lower expense ratio; Class C shares charge no front-end load and a higher expense ratio. Which investor should prefer which class, and why?

47. **[Ch.4]** An ETF trades above the value of its underlying basket. Describe what an authorized participant does and how this restores the ETF price.

48. **[Ch.4]** Why do ETF holders generally avoid the capital gains distributions that mutual fund holders receive when other investors redeem?

49. **[Ch.4]** What does a 2× leveraged ETF deliver, and why can its multi-period return differ widely from twice the index return?

50. **[Ch.4]** Where does the high distribution of a covered-call ETF come from, and what does the fund give up in return?

51. **[Ch.4]** Why must the average actively managed dollar underperform the market after costs? Use the logic of passive and active investors.

52. **[Ch.4]** Of 164 U.S. large-cap funds in the top quartile in 2021, none were still in the top quartile in 2023, while chance alone would give 6.25%. What does this suggest about choosing funds by past returns?

53. **[Ch.4]** A fund charges a 5.75% front-end load and you invest $10,000. Why must the fund earn more than 5.75% just for you to break even?

54. **[Ch.5]** Define the holding-period return (HPR), split it into its two components, and state which price is the denominator.

55. **[Ch.5]** Distinguish the arithmetic average from the geometric average of returns. Which one is the actual compound growth rate, and which is larger?

56. **[Ch.5]** How are the dollar-weighted and the time-weighted returns calculated, and when do they differ? Which one is better for judging a manager?

57. **[Ch.5]** For a given APR, how does the compounding frequency affect the effective annual rate (EAR), and how is a continuously compounded rate related to the EAR?

58. **[Ch.5]** Why does the shortcut real return ≈ nominal − inflation overstate the real return, and what is the exact relation?

59. **[Ch.5]** A bank reports a one-day 99% VaR of ₩40 million. What does the number mean, how often should it be exceeded, and is it the maximum possible loss?

60. **[Ch.5]** Under normally distributed returns, how is a one-day VaR scaled to a T-day horizon, and which common mistake does the rule avoid?

61. **[Ch.5]** As an investor changes the fraction y invested in the risky portfolio P, what happens to the Sharpe ratio of the complete portfolio, and why?

62. **[Ch.5]** How does the nominal interest rate respond to an increase in expected inflation, and what does this preserve?

63. **[Ch.5]** Explain what skew and kurtosis measure, and why they matter when returns are not normal.

64. **[Ch.5]** Compare historical simulation and Monte Carlo simulation as ways to obtain the return distribution for VaR.

65. **[Ch.5]** Why is Expected Shortfall (ES) used alongside VaR, and how did Basel treat the two?

66. **[Ch.5]** What does the Sharpe ratio measure, why is it unchanged when the investor mixes in the risk-free asset, and why is it the wrong tool for a single security?

67. **[Ch.5]** In the utility function $U = E(r) - \tfrac{1}{2}A\sigma^2$, what does A measure, and how do risk-averse, risk-neutral and risk-loving investors differ? What is a certainty equivalent?

68. **[Ch.5]** In $y^* = [E(r_p) - r_f] \div (A\sigma_P^2)$, how does the optimal fraction in the risky portfolio respond to the risk premium, the risk aversion A and the variance?

69. **[Ch.5]** Why does the capital allocation line bend (kink) at the risky portfolio P when the investor can borrow only at a rate above the risk-free rate?

70. **[Ch.5]** What is the capital market line, how is passive investing defended, and what is the Grossman-Stiglitz paradox?

71. **[Ch.6]** Distinguish systematic risk from firm-specific risk. Which can be diversified away, and which earns a risk premium?

72. **[Ch.6]** Suppose every stock has the same standard deviation and every pair has a correlation of 0.30. What happens to an equally weighted portfolio's standard deviation as n grows without limit, and what does the floor represent?

73. **[Ch.6]** In the two-asset portfolio variance formula, which term is the only one that can be negative, and why does it matter for diversification?

74. **[Ch.6]** When can two risky assets be combined into a zero-risk portfolio, and what determines the weights?

75. **[Ch.6]** Starting from 100% in a bond fund with $\sigma_B = 10\%$, when does adding a little of a riskier stock fund with $\sigma_S = 20\%$ reduce portfolio risk?

76. **[Ch.6]** Why is the minimum-variance portfolio generally not the best risky portfolio to combine with T-bills, and which portfolio is chosen instead?

77. **[Ch.6]** What does the separation property imply about the optimal risky portfolio and about the role of risk aversion? When does it break down?

78. **[Ch.6]** When does an investor hold more than 100% of her wealth in the optimal risky portfolio, and what does that mean for her financing?

79. **[Ch.6]** Why is a full Markowitz optimization hard to implement for many securities, and what must be estimated?

80. **[Ch.6]** State the insurance principle. Why is holding mostly your employer's stock in a 401(k) plan the opposite of it?

81. **[Ch.6]** Define covariance and the correlation coefficient. Why is the correlation easier to interpret, and what values can it take?

82. **[Ch.6]** In the 50/50 portfolio of a stock fund and a bond fund (standard deviations 21.73% and 3.54%), in what sense is the benefit of diversification "cost-free"?

83. **[Ch.6]** In which single case is there no benefit from diversification, and what does the opportunity set look like?

84. **[Ch.6]** On the opportunity set of two risky assets, which portfolios are efficient and which are dominated? Why?

85. **[Ch.6]** Define the efficient frontier of risky assets and how it is found. Where do the individual assets lie relative to it?

86. **[Ch.6]** Describe the four-step recipe for building the optimal complete portfolio. Which step depends on who the investor is?

87. **[Ch.6]** What is home bias, why does investing abroad lower portfolio risk, and what is the honest caveat?

Answers

1. [Ch.2] Money market instruments are **short-term** (one year or less), **highly marketable** debt. A **Treasury note** has a maturity of 1 to 10 years, so it belongs to the capital market; negotiable **certificates of deposit**, banker's acceptances and commercial paper are money market instruments.
Keywords: short-term · highly marketable · Treasury note · certificates of deposit

2. [Ch.2] The dealer **sells government securities** to an investor and agrees to **buy them back** the next day at a slightly **higher price**; the difference is the interest. Because the securities serve as **collateral**, a repo is effectively a very safe **short-term** loan to the dealer.
Keywords: sells securities · buy back · higher price · short-term · collateral

3. [Ch.2] Eurodollars are **dollar-denominated** deposits held at **foreign banks or foreign branches of U.S. banks**, outside U.S. bank regulation. The prefix refers to the **location** of the deposit, **not necessarily Europe**: a dollar deposit in a Tokyo bank is a Eurodollar deposit.
Keywords: dollar-denominated · foreign banks · location

4. [Ch.2] A pass-through is an **ownership claim** on a **pool of mortgages**; the homeowners' monthly **principal and interest** are **passed through** to investors net of servicing fees. Securitization shifts **default and prepayment risk** from the originating banks to the investors.
Keywords: ownership claim · pool of mortgages · principal and interest · passed through · default and prepayment

5. [Ch.2] Common stock is a **residual claim**: shareholders are **paid last**, after employees, suppliers and bondholders. It carries **limited liability**: the most a shareholder can lose is the **amount invested**.
Keywords: residual claim · paid last · limited liability · amount invested

6. [Ch.2] Preferred stock is a **hybrid**: like a bond it pays a **fixed dividend** and normally has **no voting rights**, but like equity the dividend is **not a legal obligation**, so skipping it does not trigger default; **cumulative** preferred must clear all arrears before any common dividend.
Keywords: hybrid · fixed dividend · not a legal obligation · no voting · cumulative

7. [Ch.2] An **ADR** is a **dollar-denominated certificate** issued by a **depositary bank** that represents ownership of **foreign shares** and trades on a **U.S. exchange**, so U.S. investors trade and receive dividends in dollars without dealing in a foreign market or currency.
Keywords: certificate · foreign shares · U.S. exchange · dollar-denominated · depositary bank

8. [Ch.2] A split lowers the stock's price, which would mechanically lower the index, so the **divisor** is reset to keep the **index level unchanged**. The split stock's **weight falls permanently**, a known distortion of price weighting.
Keywords: divisor · index level unchanged · weight falls · permanent

9. [Ch.2] It corresponds to investing an **equal dollar** amount in each stock. As prices move the weights drift, so it needs periodic **rebalancing**: **selling outperformers** and **buying underperformers**. This implies **trading costs** that value-weighted indexes do not have.
Keywords: equal dollar · rebalancing · selling outperformers · buying underperformers · trading costs

10. [Ch.2] The holder has the **right to buy** at the **exercise price**, with no obligation. If the asset ends below that price the option **expires worthless**; the **maximum loss** is the **premium** paid, while the gain is open-ended. Unlimited loss describes the call writer, not the holder.
Keywords: right to buy · exercise price · expires worthless · premium · maximum loss

11. [Ch.2] A put gives the **right to sell** at the exercise price, so it is **in the money** when the asset price is **below the exercise price**. The buyer profits only if the price falls far enough to **cover the premium** paid; the writer's gain is capped at that premium.
Keywords: right to sell · in the money · below the exercise price · premium · cover the premium

12. [Ch.2] A long **futures** position is an **obligation** to take delivery (or settle) at the agreed price, so **losses** can be large if the price falls; a long **call** is a **right** that the holder abandons if unprofitable. A call requires a **premium** at purchase, while futures require no premium, only a **margin** deposit that acts as a performance bond.
Keywords: obligation · right · premium · margin · losses

13. [Ch.2] The investor is **indifferent** when the **after-tax yields are equal**, $r_{taxable}(1 - t) = r_{muni}$, which defines a critical **marginal tax rate** t. Investors in **higher brackets** prefer the **tax-exempt** municipal bond; those below that rate prefer the taxable bond.
Keywords: indifferent · after-tax yields are equal · marginal tax rate · higher bracket · tax-exempt

14. [Ch.2] The **bank discount yield** has two flaws: "First, it assumes that the year has only **360 days**. Second, it computes the yield as a fraction of **face value** rather than of the **price the investor paid**." The **bond equivalent yield** corrects both: it uses a **365**-day year and divides by the price paid.
Keywords: bank discount yield · 360 days · face value · price the investor paid · bond equivalent yield · 365

15. [Ch.2] The **bank discount yield** is the **lowest** because it divides by face value instead of the **price paid** and uses a 360-day year. The **bond equivalent yield** corrects both and is higher. The **effective annual yield** is the **highest** because it adds **compounding** of the sub-year return, so BDY < BEY < EAY.
Keywords: bank discount yield · lowest · price paid · bond equivalent yield · effective annual yield · compounding · highest

16. [Ch.2] Money market funds try to hold their **share value at $1** and pass along all earnings as interest. A fund "breaks the buck" when losses force its value per share **below $1**. After **Lehman** Brothers went bankrupt in September 2008, the Reserve Primary Fund, which held Lehman **commercial paper**, fell to $0.97, and the resulting wave of redemptions looked like a **run** on a bank.
Keywords: share value at $1 · below $1 · Lehman · commercial paper · run

17. [Ch.2] The **principal** amount of a TIPS is **adjusted** in proportion to increases in the **Consumer Price Index**, so coupons paid on that principal rise with inflation. TIPS therefore provide a constant stream of income in **real** (inflation-adjusted) dollars.
Keywords: principal · adjusted · Consumer Price Index · real

18. [Ch.2] **General obligation bonds** are backed by the **full faith and credit** (the taxing power) of the issuer, while **revenue bonds** are issued to finance **particular projects** (airports, hospitals, turnpikes) and are backed by the **revenues from that project** or the agency operating it. Revenue bonds are **riskier** in terms of default.
Keywords: general obligation · full faith and credit · revenue bond · particular projects · revenues from that project · riskier

19. [Ch.2] A **price-weighted** index averages share prices, so a **high-priced** stock dominates it. A **value-weighted** index weights each stock by its **market capitalization**, so a **large-cap** firm dominates it. An **equally weighted** index is the simple average of each stock's return.
Keywords: price-weighted · high-priced · value-weighted · large-cap · equally weighted

20. [Ch.3] In the **primary market** a firm sells **newly issued** securities (for example an IPO) and receives the **proceeds**. Later trades between investors take place in the **secondary market** and raise no new capital, but the **liquidity** and prices it provides set the terms on which new issues can be sold.
Keywords: primary market · newly issued · proceeds · secondary market · liquidity

21. [Ch.3] In a **firm commitment** the syndicate **buys the entire issue** at a negotiated price and resells it; the spread is its pay and the **underwriter bears the risk** of a poor sale. In **best efforts** the bank only acts as an **agent** and unsold shares revert to the issuer.
Keywords: firm commitment · buys the entire issue · underwriter bears · best efforts · agent

22. [Ch.3] IPOs are on average **underpriced**: **first-day** returns are large and positive, **money left on the table** by the issuer. Explanations include compensating institutions for revealing demand in **book-building** and the **winner's curse**. Long-run returns reverse, as IPOs tend to **underperform** comparable non-issuing firms.
Keywords: underpriced · first-day · money left on the table · book-building · underperform

23. [Ch.3] The investor **buys at the asked** price ($34.90) and **sells at the bid** ($34.75). The **spread** of $0.15 is the dealer's compensation for **immediacy** and a **round-trip** transaction cost for the investor on top of commissions.
Keywords: buys at the asked · sells at the bid · spread · immediacy · round-trip

24. [Ch.3] A **market order executes immediately** at the **best available price**, so execution is certain but the price is not. A **limit order** names the **worst acceptable price**, so the price is controlled, but it **may never execute** and waits in the limit order book.
Keywords: market order · executes immediately · best available price · limit order · worst acceptable price · may never execute

25. [Ch.3] The order **cannot be filled** now; it sits in the **limit order book** and executes only if the price **falls to $47 or below** (it may expire unfilled). Such standing orders **supply liquidity** to investors who submit market orders.
Keywords: cannot be filled · limit order book · falls to $47 · supply liquidity

26. [Ch.3] The order lies **dormant** until the stock trades **at or below $55**, then becomes a **market order** to sell, which helps **limit further losses**. It does **not guarantee** the price: in a fast-falling or gapping market the sale can be filled well below $55.
Keywords: dormant · market order · at or below $55 · limit further losses · not guarantee

27. [Ch.3] High-frequency trading is **computerized** trading that exploits very **small, fleeting price discrepancies**, with positions held for **milliseconds**. Proponents say it **narrows spreads and adds liquidity**; critics note that the liquidity can be **withdrawn** under stress, as in the 2010 flash crash.
Keywords: computerized · small · milliseconds · narrows spreads · withdrawn

28. [Ch.3] Margin **magnifies both gains and losses** on the investor's equity because the loan is fixed while the collateral value moves. **Interest accrues** whatever the stock does, so the stock must earn more than the **borrowing rate** for leverage to pay. The initial margin requirement is set by the **Federal Reserve**, and brokers may be stricter.
Keywords: magnifies · gains and losses · interest accrues · borrowing rate · Federal Reserve

29. [Ch.3] For a **long** margin purchase the loan is fixed, so a **price fall** cuts equity below the **maintenance margin**; for a **short** position the liability grows when the **price rises**, so the call comes on a rise. In both cases the broker demands **additional funds** or sells shares.
Keywords: long · price fall · short · price rises · maintenance margin · additional funds

30. [Ch.3] The short seller **borrows** shares and **sells** them, then **buys back** later to return them, profiting if the **price falls**. The gain is **capped** because the price cannot fall below zero, while the loss is **unlimited** because the price can rise without bound.
Keywords: borrows · sells · buys back · price falls · capped · unlimited

31. [Ch.3] The sale **proceeds stay in the account as collateral**, and the short seller must post **additional margin**. The lender keeps the economic rights of ownership, so the short seller must **pay the lender any dividends** the stock pays, a **carrying cost** of the trade.
Keywords: collateral · additional margin · dividends · pay the lender · carrying cost

32. [Ch.3] ECNs (electronic communication networks) match buy and sell orders directly with **no dealer** in the middle. **Algorithms** decide the timing, price and size of orders to cut cost and market impact, and ultra-fast high-frequency programs exploit tiny price gaps in **milliseconds**. The benefits are faster execution, **lower trading costs**, greater price transparency, anonymity for large orders and trading beyond exchange hours.
Keywords: ECN · no dealer · algorithm · milliseconds · lower trading costs

33. [Ch.3] The **Securities Acts of 1933** and 1934 require **full disclosure** at issue and created the **SEC** to oversee markets. **Self-regulation (FINRA)** means industry bodies police member conduct and trading practices. **Sarbanes-Oxley** (2002) tightened corporate governance, auditing and financial reporting after major scandals. **Insider-trading** rules prohibit trading on **material non-public information**.
Keywords: 1933 · full disclosure · SEC · FINRA · Sarbanes-Oxley · material non-public information

34. [Ch.4] She owns a **proportional (pro-rata) claim** on the **whole pooled portfolio**, 2% of every asset, and receives 2% of its **income**. She does **not own specific securities**, and no return is **guaranteed**.
Keywords: proportional · whole pooled portfolio · income · specific securities

35. [Ch.4] The four services are **record keeping and administration**, **diversification and divisibility**, **professional management**, and lower **transaction costs** from trading in large blocks. A **guaranteed** return above the market is not among them; whether management beats the market is an empirical question.
Keywords: record keeping · diversification · professional management · transaction costs · guaranteed

36. [Ch.4] NAV = (**market value of assets** − **liabilities**, such as accrued management fees) ÷ **shares outstanding**. New shares are issued **at NAV**, so assets and shares rise in **proportion** and NAV is unchanged, with **no dilution** of existing holders.
Keywords: market value of assets · liabilities · shares outstanding · at NAV · proportion

37. [Ch.4] It is filled at the **NAV struck once a day at the 4:00 p.m. close**, the **same price** for every order that day, and the investor does **not know** it when ordering. Closed-end funds and ETFs trade on an **exchange** at intraday market prices.
Keywords: NAV · 4:00 p.m. · same price · not know · exchange

38. [Ch.4] A closed-end fund has a **fixed number** of shares after its **IPO** and neither issues nor redeems them. Investors trade with each other on an **exchange**, so the price is set by **supply and demand** and can sit at a **premium or discount** to NAV.
Keywords: fixed number · IPO · exchange · supply and demand · premium or discount

39. [Ch.4] A UIT is a **fixed, unmanaged** portfolio deposited in a trust by a **sponsor** and sold as redeemable units; it **terminates on a set date**. With essentially **no trading** after formation, **management fees are very low**.
Keywords: fixed · terminates · sponsor · no trading · very low

40. [Ch.4] Hedge funds are private pools open only to **accredited and institutional** investors and largely **exempt from the Investment Company Act** of 1940. They use **short sales, derivatives and heavy leverage**, charge higher fees (**2 and 20**), and restrict redemptions through **lock-ups**.
Keywords: accredited · Investment Company Act · short sales · 2 and 20 · lock-up

41. [Ch.4] A **balanced fund** holds **stocks and bonds** in a relatively **stable mix**; an **asset allocation fund** actively **shifts** the mix across asset classes as market conditions change.
Keywords: balanced fund · stocks and bonds · stable mix · asset allocation fund · shifts

42. [Ch.4] Operating expenses are **deducted daily from fund assets** before NAV is struck, so the investor is **never billed** separately and the cost is easy to overlook. The reported return is already **net of** the expense ratio.
Keywords: deducted · daily · from fund assets · never billed · net of

43. [Ch.4] A **12b-1 fee** is an **annual** charge of up to **1%** of assets that pays for **marketing and distribution** (including adviser compensation). It is reported inside the **expense ratio**, but economically it is a **load** paid in instalments.
Keywords: annual · marketing and distribution · 1% · expense ratio · load

44. [Ch.4] A **CDSC** is charged on redemption and **declines** each year until it disappears. It makes **early redemption** expensive and so **locks investors in**; it compensates the **selling broker**, not the fund.
Keywords: CDSC · declines · early redemption · locks investors in · selling broker

45. [Ch.4] No: the reported return **overstates** her return. Published returns are computed from **NAV**, which is net of the **expense ratio** but **not of loads**; the load **reduces the amount invested**, so over one year a 5% load can consume almost half of an 11% return.
Keywords: overstates · not of loads · NAV · expense ratio · reduces the amount invested

46. [Ch.4] The load is a **one-time** cost but the higher expense ratio is paid **every year** and compounds. Over a **short** horizon the load dominates, so **Class C** wins; over a **long** horizon the annual fee dominates, so **Class A** wins.
Keywords: Class A · Class C · long · short · one-time · every year

47. [Ch.4] The **authorized participant buys the cheaper basket**, swaps it **in kind** for newly **created** ETF shares, and **sells the ETF shares** on the exchange. The selling pressure pushes the ETF price **back toward the basket value**.
Keywords: authorized participant · buys the cheaper basket · in kind · sells the ETF shares · back toward the basket value

48. [Ch.4] In a **mutual fund**, **redemptions force** the fund to **sell stock**, realizing gains that are distributed to all holders. In an **ETF** the seller finds a buyer on the **exchange**, and creations and redemptions happen **in kind**, not for cash, so the fund does not need to sell.
Keywords: mutual fund · redemptions force · sell stock · ETF · exchange · in kind

49. [Ch.4] It aims for **twice the index's daily return**; the fund **resets** to 2× every day, so gains and losses **compound** on a changing base. In **volatile** markets it falls behind even if the index ends flat, so multi-period returns are not 2×.
Keywords: daily · twice · resets · compound · volatile

50. [Ch.4] The fund **sells call options** on shares it owns and distributes the **premium** received. That premium is the price of the **upside it gives away** (gains are capped above the strike) while the **downside remains open**; it is an option strategy, **not a dividend** and not insurance.
Keywords: sells call options · premium · upside · downside · not a dividend

51. [Ch.4] Passive investors hold the market, so what remains for **active investors as a group** is also the market: their value-weighted average **gross return** equals the market return. After **fees** the average active dollar must **lag** the market. This says nothing about any single fund.

52. [Ch.4] Past top performance is **not a reliable** signal of future top performance: if **skill** drove the rankings, winners would repeat more often than **chance**, not less. The evidence shows no **persistence**; the lasting signal is the **fee**, not past returns.
Keywords: persistence · not a reliable · chance · skill · fee

53. [Ch.4] A front-end **load** is deducted before investing, so a 5.75% load on $10,000 leaves only **$9,425** as the **net** amount invested. The load is 5.75% of the gross, but to get back to $10,000 you must earn **6.10%** on the net, i.e. you need more than the load percentage just to **break even**.
Keywords: load · net · 9,425 · 6.10% · break even

54. [Ch.5] HPR = (ending price − beginning price + dividends) ÷ beginning price, the **capital gain yield** plus the **dividend yield**. The denominator is always the **purchase price**, **never the ending price**.
Keywords: capital gain yield · dividend yield · purchase price · never the ending price

55. [Ch.5] The **arithmetic** average is the simple mean of period returns; the **geometric** average is the constant **compound** growth rate that reproduces the actual ending wealth. The geometric average is **smaller** than the arithmetic one when returns fluctuate, by roughly $\frac{1}{2}\sigma^2$, and the gap grows with **volatility**.
Keywords: arithmetic · geometric · compound · smaller · volatility

56. [Ch.5] The **dollar-weighted** return is the **IRR** of all **cash flows** (purchases, dividends, sales); the time-weighted return is the **geometric** average of **sub-period** returns. They **differ** when money is added or withdrawn at different times, and the time-weighted figure ignores that timing, so it is better for judging the **manager**.
Keywords: IRR · cash flows · geometric · differ · manager

57. [Ch.5] For a given APR, **more frequent** compounding gives a **higher EAR**, $\text{EAR} = (1 + \text{APR}/m)^m - 1$. With **continuous** compounding the APR equals the **natural log** of (1 + EAR).
Keywords: EAR · more frequent · higher · continuous · natural log

58. [Ch.5] The exact relation is the **Fisher** equation, $1 + r = (1 + R) \div (1 + i)$. The shortcut $R - i$ is an **approximation** that **overstates** the real return by the **cross term** $r \times i$, and the error grows when inflation is high.
Keywords: Fisher · approximation · overstates · cross term

59. [Ch.5] VaR is the loss that is **exceeded** with a given **probability** over the horizon; here the loss should exceed ₩40 million on about **1 trading day in 100**. It is **not the maximum** loss: on the worst days losses can be **larger**, and VaR says nothing about how much larger.
Keywords: probability · exceeded · 1 trading day in 100 · not the maximum · larger

60. [Ch.5] Under normal returns $\mathbf{VaR = V_0 \times z \times \sigma \times \sqrt{T}}$: the one-day figure is scaled by the **square root** of time because independent daily variances add, **not multiplied by T**, which would overstate the risk. The **z-value** reflects the **confidence level** (1.645 for 95%, 2.326 for 99%) and σ is the daily **volatility**.
Keywords: square root · not multiplied by T · z-value · volatility

61. [Ch.5] It is **unchanged** for every y: both the **risk premium** and the **standard deviation** of the complete portfolio **scale** with y, so their ratio equals the **slope** of the capital allocation line. This is why the choice of P and the choice of y can be solved separately.
Keywords: unchanged · risk premium · standard deviation · scale · slope

62. [Ch.5] Investors need compensation for expected inflation, so the **nominal** rate **rises** with **expected inflation**: $r_{nom} = r_{real} + E(i)$. The increase in the nominal rate is needed to **preserve** investors' **real** rate of return.
Keywords: expected inflation · nominal · real · rises · preserve

63. [Ch.5] Skew measures the **asymmetry** of the return distribution. **Kurtosis** compares the frequency of **extreme** values with that of the **normal distribution** (whose kurtosis is zero), so it is called a measure of **fat tails**, which signal tail risk that the standard deviation alone understates.
Keywords: skew · asymmetr · kurtosis · fat tails · extreme · normal distribution

64. [Ch.5] Historical simulation sorts the **actual past returns** and reads off the quantile; **no distribution** is assumed so **fat tails** come for free, but it needs a long sample. **Monte Carlo** simulation draws many paths from an **assumed model**, handles **options** and other non-linear payoffs, but carries model risk.
Keywords: historical simulation · actual past returns · no distribution · fat tails · Monte Carlo · assumed model · options

65. [Ch.5] VaR marks the **cut-off** but says nothing about how bad the loss is beyond it. ES (also CVaR) is the **average loss** in the tail **beyond** VaR. ES is **coherent** (subadditive) where VaR is not, so merging books cannot manufacture risk, and **Basel FRTB** replaced the 99% VaR capital charge with **97.5%** expected shortfall.
Keywords: cut-off · average loss · beyond · coherent · Basel · 97.5%

66. [Ch.5] The Sharpe ratio is the **reward per unit** of volatility, $[E(r_P) - r_f] / \sigma_P$. It equals the **slope of the CAL** and is **unchanged** however much of the risk-free asset is mixed in, so it measures the quality of the risky portfolio itself. It is the wrong tool for a **single security** because the denominator includes **diversifiable risk**.
Keywords: reward per unit · slope of the CAL · unchanged · single security · diversifiable risk

67. [Ch.5] A **risk-averse** investor has $A > 0$, a **risk-neutral** investor has $A = 0$ and judges by E(r) alone, and a **risk-loving** investor has $A < 0$. The term $\frac{1}{2}A\sigma^2$ is the **penalty** for risk. The **certainty equivalent** is the rate that a risk-free investment would have to offer to give the same utility as the risky portfolio.
Keywords: risk-averse · risk-neutral · risk-loving · penalty · certainty equivalent

68. [Ch.5] y^* **maximizes utility** and **rises with the risk premium**, **falls with risk aversion** and **falls with variance**: a larger reward per unit of risk raises the risky share, while a larger risk penalty lowers it.
Keywords: maximizes utility · rises with the risk premium · falls with risk aversion · falls with variance

69. [Ch.5] To the left of P the investor **lends** at the risk-free rate. Beyond P she must borrow at a rate above r_f, and because the **borrowing rate is higher** the **slope falls** (the Sharpe ratio of leveraged positions is lower), so the CAL has a **kink** at P.
Keywords: lends · borrowing rate is higher · slope falls · kink

70. [Ch.5] When a **broad market index** is used as the risky portfolio P, the resulting capital allocation line is the **capital market line** (CML). Passive investing is defended by the **free-rider** argument (the analysis of thousands of other investors is already in the price) and by **low cost** (index funds charge about 0.03-0.2% a year against 1% or more for active funds). The open question is **who sets prices** if everyone is passive, the **Grossman-Stiglitz** paradox.
Keywords: capital market line · broad market index · free-rider · low cost · who sets prices · Grossman

71. [Ch.6] Total variance is **systematic** plus **firm-specific** variance. Adding stocks shrinks **firm-specific** (diversifiable) risk toward zero but leaves systematic (market) risk. Prices are set by **diversified investors**, so only the systematic part earns a **risk premium**.
Keywords: systematic · firm-specific · diversified investors · risk premium

72. [Ch.6] As n grows the firm-specific term $(1/n)\sigma^2$ vanishes and σ_P **falls toward a floor** equal to the square root of the **average covariance**. The floor is **market (systematic) risk**, which **cannot be diversified** away.
Keywords: falls · floor · average covariance · market risk · cannot be diversified

73. [Ch.6] The **interaction (covariance) term** $2w_S w_B \cdot \text{Cov}$ is the only one that can be **negative**; the other two terms are **squares** and always positive. When the **correlation** is below +1 it shrinks the variance, and for $\rho < 0$ it pulls σ_P below the **weighted average** of the two standard deviations.
Keywords: interaction term · negative · correlation · weighted average · squares

74. [Ch.6] When the assets are **perfectly negatively correlated** ($\rho = -1$) a **zero-risk** portfolio exists: setting $\sigma_P = |w_S\sigma_S - w_B\sigma_B| = 0$ gives $w_S = \sigma_B \div (\sigma_S + \sigma_B)$, so the weights are set by the **standard deviations**.
Keywords: perfect negative correlation · zero risk · standard deviations

75. [Ch.6] Risk falls when the **correlation** is **below $\sigma_B \div \sigma_S$** (here $10 \div 20 = \mathbf{0.5}$), the point at which the minimum-variance stock weight becomes positive. A negative correlation is **not required**.
Keywords: correlation · below $\sigma_B \div \sigma_S$ · 0.5 · not required

76. [Ch.6] The minimum-variance portfolio only minimizes risk and **ignores expected return**. The optimal risky portfolio is the **tangency** point where the **capital allocation line** is steepest, so it maximizes the **Sharpe ratio**, which is **at least as high** as that of any other mix, including the minimum-variance portfolio.
Keywords: ignores expected return · Sharpe ratio · tangency · capital allocation line · at least as high

77. [Ch.6] One optimal risky portfolio is the same for every investor who shares the same inputs; only the **split** between it and **T-bills** depends on **risk aversion**. The property breaks down with **different inputs, borrowing rates above lending rates, taxes** and human capital.
Keywords: same for every investor · risk aversion · split · T-bills · different inputs

78. [Ch.6] A **risk-tolerant** (low risk aversion) investor gets $y^* = [E(r_P) - r_f] \div (A\sigma_P^2)$ **above 1**, so she **borrows** at the **risk-free rate** and invests more than her wealth in P, a **leveraged** position.
Keywords: low risk aversion · above 1 · borrows · risk-free rate · leveraged

79. [Ch.6] With n securities you must estimate **expected returns** for each, **n variances** and $n(n-1)/2$ **covariances**, a total of $n(n+3)/2$ that **grows with the square of n** (125,750 inputs for n = 500). Every estimate carries **estimation error**.
Keywords: expected returns · n variances · covariances · square of n · estimation error

80. [Ch.6] "Risk reduction by **spreading exposure** across many **independent risk sources** is called the **insurance principle**." An employee who also holds company stock in her 401(k) is **doubling down** on the fortunes of a **single firm**, since her job and her savings depend on the same firm.
Keywords: insurance principle · spreading exposure · independent risk sources · single firm · doubling down

81. [Ch.6] Covariance measures how much two returns move together, but its units are "**percent square**," so its size is hard to interpret. The **correlation coefficient** rescales it by the **product of the standard deviations**, $\rho = \text{Cov} \div (\sigma_S\sigma_B)$, so $\text{Cov} = \rho\sigma_S\sigma_B$, and it lies **between −1 and +1**.
Keywords: covariance · percent square · correlation coefficient · product of the standard deviations · between −1 and +1

82. [Ch.6] The portfolio's expected return (8.00%) is exactly the **weighted average** of the two funds' returns, so you **lost nothing** in return. Yet the portfolio standard deviation (9.57%) is far **below** the weighted average of the two (12.64%). That 3.07-point gap is the **diversification benefit**, and it exists purely because the **correlation** is below +1.
Keywords: weighted average · lost nothing · below · correlation · diversification benefit

83. [Ch.6] Perfect positive correlation ($\rho = +1$) is the **only** case with no benefit from diversification: $\sigma_P = w_B\sigma_B + w_S\sigma_S$ is a **weighted average** of the component standard deviations and the opportunity set is a **straight line** through the two assets. Whenever returns are **less than perfectly** positively correlated, σ_P is below that weighted average.
Keywords: perfect positive correlation · only · weighted average · straight line · less than perfectly

84. [Ch.6] The **upward-sloping** part of the opportunity set, starting at the **minimum-variance portfolio**, is the **efficient** part. Points below it are **dominated**, because a portfolio on the efficient part offers a **higher expected return** at the same standard deviation.
Keywords: efficient · upward-sloping · minimum-variance portfolio · dominated · higher expected return

85. [Ch.6] The **efficient frontier** is the set of portfolios that offer the **highest expected return** for each level of risk. It is found by **minimizing variance** subject to each **target expected return**. Individual assets all lie **inside** the frontier, so every one of them is dominated.
Keywords: efficient frontier · highest expected return · minimizing variance · target expected return · inside

86. [Ch.6] Step 1 collects the **inputs**: expected returns, standard deviations and covariances or correlations. Step 2 traces the **efficient frontier** (upper branch). Step 3 **maximizes the Sharpe ratio** to find the optimal risky portfolio P*. Step 4 sets **y*** from the investor's **risk aversion** and splits wealth between P* and the risk-free asset. Only step 4 depends on the investor; steps 1-3 are identical for everyone, the **separation property**.
Keywords: inputs · efficient frontier · Sharpe ratio · y* · separation property

87. [Ch.6] Investors everywhere **overweight** their home country relative to its share of the world portfolio, a pattern called **home bias**. Adding foreign assets lowers the **covariance** term in the portfolio variance because countries' systematic shocks differ. The caveat is that global **correlations** have risen and **currency** risk adds another layer, yet the benefit is not zero as long as correlations stay **below +1**.
Keywords: home bias · overweight · covariance · correlations · below +1 · currency